Numerical Computation — Test 1

Reconstructed Questions

Note. I reconstructed these questions from memory, and the wording was prepared with the help of generative AI (GenAI). This is not an official version of the test. The wording may differ from the original, and some numbers, details or parts of the questions might be incorrect or missing.

1. Householder reflections and QR factorisation

(i) Write down the definition of Householder reflections and use them to compute the QR factorisation of the matrix

$$A = \begin{pmatrix} 9 & -11 \\ 12 & 2 \end{pmatrix}.$$

(ii) Is the QR factorisation unique? If not, what condition on R makes it unique?

(iii) For which kind of matrices does one prefer QR factorisation over LU factorisation?

2. Givens rotations

Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 1 & 3 \end{pmatrix}.$$

Compute the Givens rotations G_1, G_2 such that

$$G_2 G_1 A = \begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & * & * \end{pmatrix}.$$

3. Least squares and the pseudoinverse

Let $A \in \mathbb{R}^{n \times n}$ and $b \in \mathbb{R}^n$ be given.

(i) Show that every minimiser x^* of the minimisation problem

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|_2 \tag{1}$$

satisfies

$$A^\top A x^* = A^\top b.$$

(ii) If A is injective and its singular value decomposition is given, write down the formula for the pseudoinverse of A.

(iii) The pseudoinverse is defined by $A^\dagger b := x^*$, where x^* is the minimal-norm solution to (1). Compute $A^\dagger e_1$ and $A^\dagger e_2$ to find the pseudoinverse of

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

4. Singular value decomposition

(i) Give the definition of the singular value decomposition of a matrix $A \in \mathbb{R}^{n\times n}$.

(ii) Compute the singular value decomposition of

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

Check whether your decomposition is correct.

(iii) How can the singular value decomposition be used for image compression?

5. Krylov spaces, conjugate gradients and Newton's method

(i) Write down the definition of the Krylov space $\mathcal{K}_\ell(A, x)$ for a matrix A, a vector x and $\ell \in \mathbb{N}$.

(ii) For which kind of matrices A is the conjugate gradient method suitable?

(iii) How do you compute a step in Newton's method in $\mathbb{R}^n$? Write down the formula. What changes with the damped Newton method?